

Fig. 2: An actual-size, single-side PCB for the microcontroller-based scientific calculator

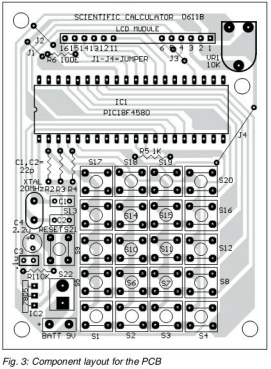


Fig. 3: Component layout for the PCB

Microcontroller PIC18F4580 is at the heart of this calculator. It has 32 kB of enhanced Flash programmable memory, five bidirectional input/output ports, four timers, 11-bit analogue-to-digital converter, two comparators and is also capable of serial communication. It offers different oscillator options to provide the basic clock frequency.

Power-on reset is provided by the combination of resistor R1 and capacitor C4. Switch S21 is used for manual reset. A 20MHz crystal connected to pins 13 and 14 provides the basic clock to the microcontroller. Port pins RD4 through RD7 of the microcontroller are connected to data port pins D4 through D7 of the LCD module, respectively.

The microcontroller drives the LCD in 4-bit mode and sends the data in two nibbles (4-bit parts) for display on the LCD. Port pins RC0, RC1 and RC5 are connected to read/write ($R/\overline{W}$), register-select (RS) and enable (E) of the LCD, respectively, to control the LCD operation. Preset VR1 is used for contrast control of the LCD.

The 5×4-matrix keypad is interfaced to microcontroller PIC18F4580 through its port pins. Its five rows are connected to five pins of Port B (RB0 through RB4) and four columns to four pins of Port A (RA0 through RA3). Key debounce technique is used to eliminate false key-press detection. The microcontroller scans the keypad continuously to detect and identify the key pressed by the user. Eight out of 20 keys (S1 through S20 switches) can perform more than one function at a time.

IC 7805 (IC2) provides regulated 5V to the circuit. Capacitor C3 bypasses ripples, if any, in the regulated supply.

An actual-size, single-side PCB for the microcontroller-based scientific calculator is shown in Fig. 2 and its component layout in Fig. 3. Assembling the circuit on a PCB minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Use bases to avoid damage to ICs due to direct soldering and overheating.

Keypad Details

Switch	Function
S1	1
S2	2
S3	3
S4	4
S5	5
S6	6
S7	7
S8	8
S9	9
S10	0
S11	+/-
S12	AC
S13	ADD/SUB
S14	MUL/DIV
S15	POW/SQR/SQRT
S16	INV/EXP
S17	EXC
S18	LOG/LN
S19	SIN/COS/TAN
S20	ASIN/ACOS/ATAN

Functions of different keys

Functions assigned to different switches on the keypad are shown in the table. Fig. 4 shows the user manual of the calculator for division of two numbers.

The '+/–' key is used to give sign to the number. It works only when the cursor is below letter 'S' displayed on the LCD. If it is pressed once '+' sign is displayed, and if it is pressed twice '-' sign is displayed.

Digits 0 through 9 on the keypad have the same functions as in a normal calculator. You can enter digits only when the cursor is below digit '0' displayed in the LCD. When you press a number key, '0' displayed on the LCD changes to the number pressed.

After the first data is inputted, the calculator shows word 'operator' in the second line of the LCD. There are seven operator keys. When an operator